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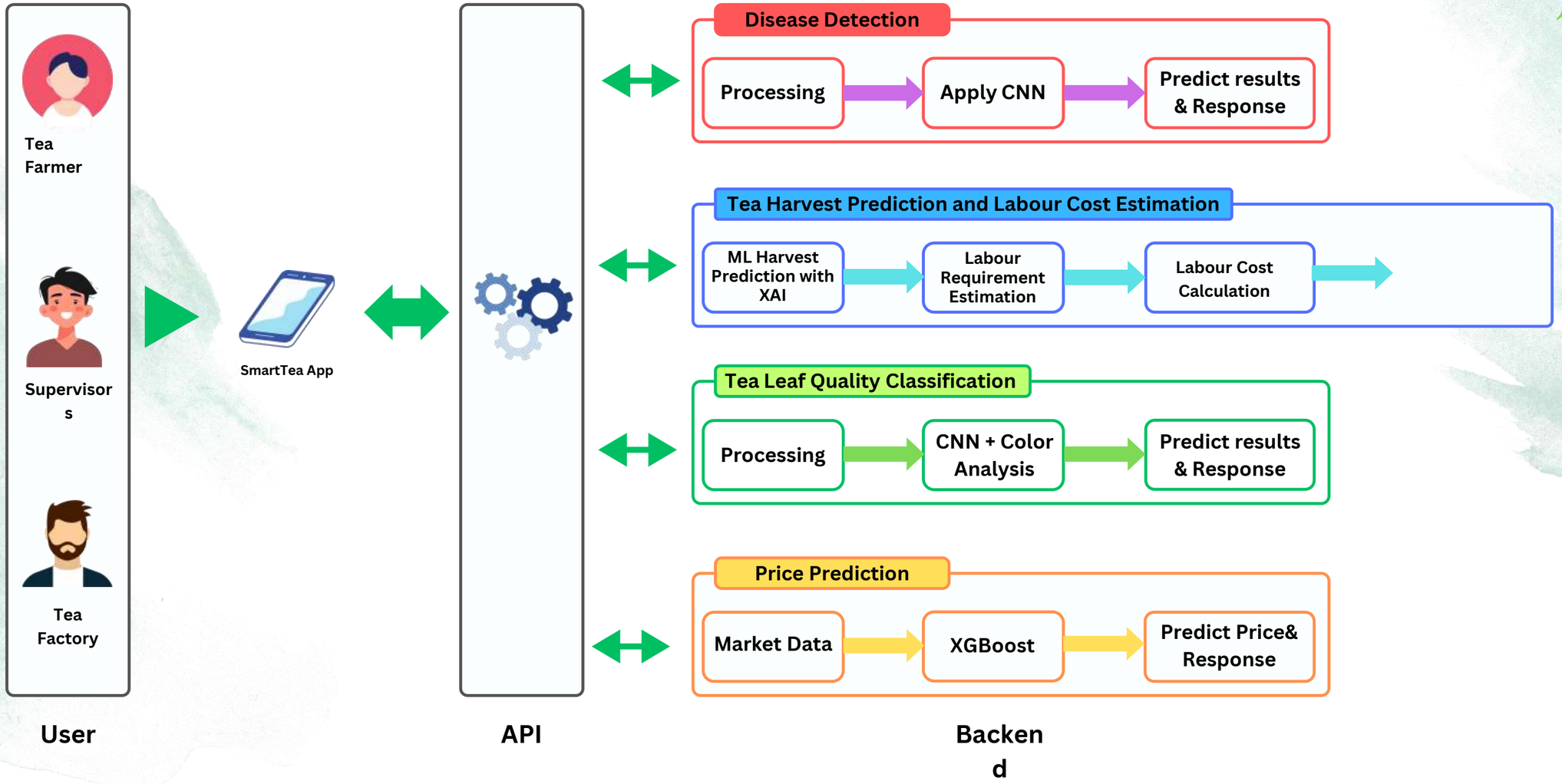
GROUP-25-26J-343

Overall System Diagram



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Tea Leaf Disease Detection

Specialization: Information Technology

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09/09/2025

Tea Leaf Disease Detection

Problem:

- Farmers manually check for Grey & Brown Blight which is slow, subjective and error prone.
- Lack of low cost automated solutions for real time field detection.

Technologies and Methods	Research Paper [1]	Research Paper [2]	Research Paper [3]	Research Paper [4]	
Grey & Brown Blight detection	✅	✅	❌	✅	❌
Mobile/field real-time usability	❌	❌	❌	❌	❌
Offline inference (no internet needed)	❌	❌	❌	❌	❌
Dataset localized to	❌	✅	❌	❌	✅

Tea Leaf Disease Detection

Key Domains of Specialization-based Knowledge

- Machine Learning & Deep Learning – for image based classification.
- Image Processing – for preprocessing, segmentation and feature extraction.
- Mobile Application Development – for deploying real time disease detection on smartphones.
- Data Science & Model Evaluation – for dataset preparation, augmentation and performance analysis.

Latest Technologies Used

- Convolutional Neural Networks (CNNs) – MobileNetV2 for lightweight yet accurate classification.
- Transfer Learning – leveraging pre trained models to improve accuracy with limited datasets.
- OpenCV – for preprocessing (resizing, normalization).
- TensorFlow Lite – optimized model deployment on mobile devices.
- Segment Anything Model (SAM) + CNN – for advanced segmentation and early disease spot detection.

Tea Leaf Disease Detection

Evaluation & Success Metrics

- Validation Methods: Test split, Cross validation.
- Performance Metrics: Accuracy, Precision, Recall, F1-score, Confusion Matrix.
- Expected Results: >95% classification accuracy for Grey Blight & Brown Blight.
- Success Proof: Real field testing with Sri Lankan tea leaves to confirm robustness.

Data Availability & Ethical Clearance

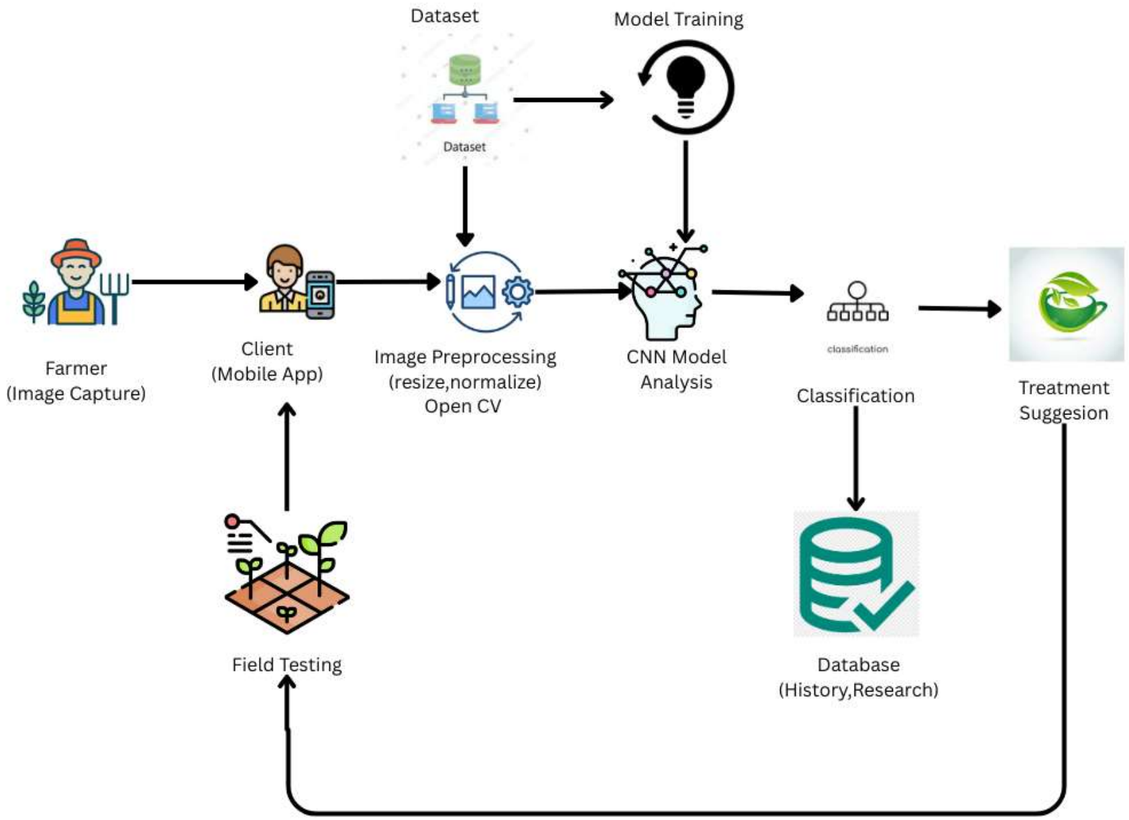
Data Sources:

- Kaggle Tea Leaf Disease Dataset (includes Grey & Brown Blight).
- Field datasets from research papers (Bangladesh, China, Taiwan).
- Self collected images from Sri Lankan plantations (future dataset expansion).
- Data Augmentation: Rotation, zoom, color adjustments to balance dataset.

Ethical Considerations:

- Collected data will be used only for research & non commercial purposes initially.
- Proper permissions from tea estates & farmers.
- No personal/identifiable data collected – only leaf images.

Tea Leaf Disease Detection



Tea Leaf Disease Detection

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Investment / Cost: Labor :

- (Developers, AI engineers) and materials (cloud, devices) required for building and deploying the system.

Recovery / Pricing :

- Revenue through subscription fees and one time setup charges to recover costs.

Target Market / Customer Profile :

- Tea estates, agronomists, and research institutes who need data driven disease management.

Value Proposition :

- Provides early disease detection, reduces manual inspection effort, and improves crop yield.
- **Intellectual Property Rights:** Software/AI patents, platform trademark, and licensing of collected data.

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Harvest Prediction & Labour Cost Estimation

Specialization: Information Technology

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Harvest Prediction & Labour Cost Estimatiom

Problem Definition

- Yield and labour cost estimation are often manual and inaccurate.
- Existing models rely on costly IoT devices not suitable for small farmers.

Research Gap

Feature	[1] Crop Yield ML Review	[2] Crop Yield ML	[3] Indian Agri ML	[4] Spectral Yield Estimation	[5] SAR Biomass Estimation	Proposed Solution
Low-Cost / IoT-Free	✗	✗	✗	✗	✗	✓
Labour Requirement Estimation base on predicted harvest	✗	✗	✗	✗	✗	✓
Cost Estimation	✗	✗	✗	✗	✗	✓
Explainable Predictions (XAI)	✗	✗	✓	✗	✗	✓

Harvest Prediction & Labour Cost Estimatiom

Key Domains Used

- Data Science -(process the tea and weather data)
- Artificial Intelligence & Machine Learning-(predict yield,)
- Agricultural Economics for labour cost modeling-(calculate labour cost,)
- Explainable AI (XAI) for transparent predictions-(like rainfall, fertilizer, or plant
- age.)

Latest Technologies Used

Main programming

Explain the model

collaboratiion

ML algorithm

training and deploying

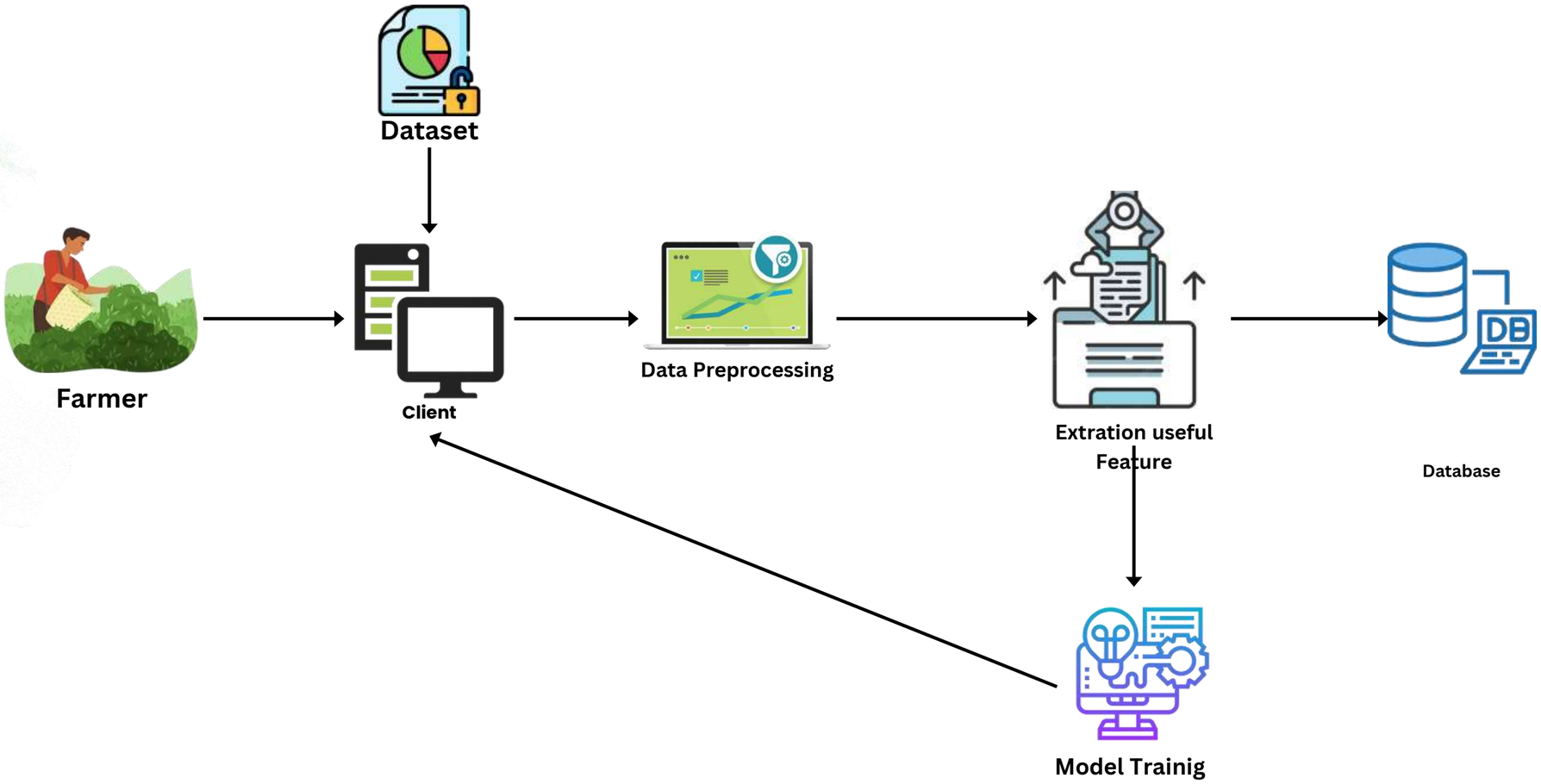
Harvest Prediction & Labour Cost Estimatiom

Evaluation Plan

- Avoid Data Leakage.
- Compare with Baseline Models
- High accuracy.
- High recall.
- Avoid False Predictions
- Practical Validation

Data availability and Ethical clearance

Harvest Prediction & Labour Cost Estimatiom



So in short – the farmer will know how much tea to expect, how many workers are needed, and how much it will cost – all before the harvest begins."

References

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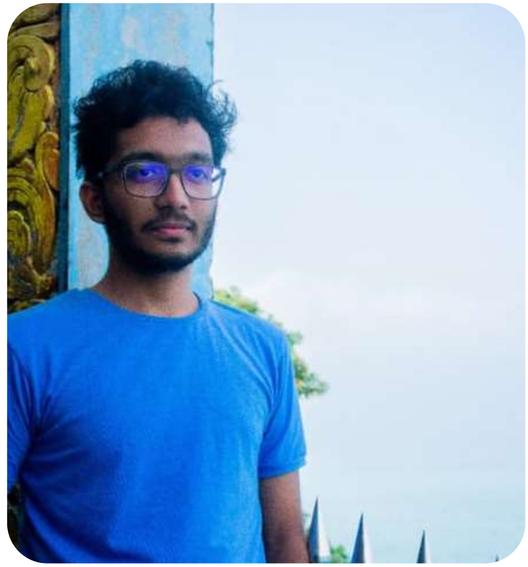

Tea Leaf Quality Classification

Specialization: Information Technology

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Tea Leaf Quality Classification

- Tea factories in Sri Lanka manually inspect tea leaves by sorting samples into Best, Below Best, and Bad.
- Manual process is time-consuming, subjective, and prone to human error.
- No widely adopted automated, low-cost system exists for real-time tea leaf quality assessment.

Technologies and Methods	Research Paper [1]	Research Paper [2]	Research Paper [3]	Research Paper [4]	Research Paper [5]	Proposed Method
Bulk tea leaf evaluation	✓	✗	✗	✓	✗	✓
Fresh leaf classification (factory intake)	✗	✓	✓	✓	✓	✓
Mobile/field-level real-time usability	✗	✗	✗	✗	✗	✓
Dataset localized to Sri Lankan standards	✓	✗	✗	✗	✗	✓
Explainable AI for transparent classification	✗	✗	✗	✗	✗	✓
IoT-free deployment	✗	✗	✗	✗	✗	✓

Creative Solution: A mobile AI application that uses a phone camera to scan tea leaf samples, classify into Best / Below Best / Bad, calculate percentages, and instantly decide accept/reject.

Tea Leaf Quality Classification

Key Domains

- **Artificial Intelligence & Machine Learning** – Deep learning for tea leaf image classification.
- **Computer Vision** – Image processing to detect and separate leaves in samples.
- **Mobile Application Development** – Deploying AI model into a smartphone app.
- **Data Science** – Dataset collection, labeling, and accuracy validation.

React Native

Technologies

- **Deep Learning Model:** **Vision Transformers (ViT & Swin Transformer)** → High accuracy for image classification, outperforming CNNs.
- **Framework:** **PyTorch** → Modern deep learning framework with strong support for Transformers.
- **Mobile Platform:** **Fluttere**→ Cross-platform (Android + iOS) with smooth AI model integration.
- **Cloud/Edge Deployment:** **TensorFlow Lite** → Optimized AI inference directly on mobile devices

Tea Leaf Quality Classification

Evaluate Plan

- How to Evaluate / Prove Success
- Accuracy Metrics: Precision, Recall, F1-score of classification (Best / Below Best / Bad).
- Comparison: Automated system vs. manual inspection by tea experts.
- Performance Measures: Processing time per sample, user satisfaction (survey).
- Expected Accuracy: 85% – 95% (based on prior studies in plant/tea leaf classification).

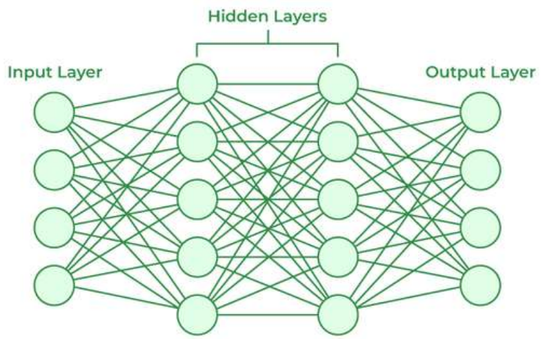
Data availability and Ethical clearance

- **Data Source:** Tea leaf images collected from Sri Lankan tea estates/factories.
- **Dataset Preparation:** Proper labeling into Best / Below Best / Bad by experts.
- **Ethical Clearance:**
  - No personal data involved → low ethical risk.
  - Data collection will be done with factory/estate permission.
  - Images will be used only for research and kept confidential.

Tea Leaf Quality Classification

Input Image

Image Patches



Linear Projection + Position Embedding

Transformer Encoder Layers

Class Token Output

Classification Head

Accept or Reject

Tea Leaf Quality Classification

1. Factory worker places a **random sample of tea leaves** on a white background.

2. Opens the **mobile app** and scans the sample using the phone camera.

3. AI processes the image and displays **classification results + percentages.**

4. Factory manager gets an **instant Accept/Reject** decision, saving time and labor.

Tea Leaf Quality Classification

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AI - Driven Price Forecasting Grade Specific Ceylon Tea

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09/09/2025

AI - Driven Price Forecasting Grade Specific Ceylon Tea

- **Export planning relies on historical data that mismatches in supply and demand.**
- **Existing systems that focus only on auction price , no grade/country-specific forecasts.**
- **Shortcomings: Ignore global demand, shipping cost , no mobile access.**

Technologies and Methods	Research Paper [1]	Research Paper [2]	Research Paper [3]	Research Paper [4]	Research Paper [5]
Short term tea price forecasting	✓	✓	✓	✓	✓
Grade specific forecasting	✗	✗	✗	✓	✗
Export volume forecasting	✗	✓	✗	✗	✗
Country specific demand analysis	✗	✗	✓	✗	✗
AI/ Deep Learning	✗	✓	✓	✗	✗

Proven Gap

There is no existing solution that combines SmartTea AI system forecasting monthly exports by grade & country via mobile app.

AI - Driven Price Forecasting Grade Specific Ceylon Tea

Key Domains of Specialization Knowledge.

- Data Science - Handling data.
- Artificial Intelligence & Machine Learning : Deep learning, ANN, ARIMA, LSTM, Hybrid for forecasting.
- Flask/Django backend.
- React Native/ Flutter - For the frontend (mobile app).

Latest Technologies I Used

- ARIMA, LSTM, - To capture linear and non linear patterns.
- Tensor Flow/ Keras, Scikit - To implement deep learning model.
- Hybrid DB - Store both structured and unstructured data.

AI - Driven Price Forecasting Grade Specific Ceylon Tea

Application of Specialization - based Knowledge

Evaluation

Metrics RMSE, MAE, MAPE, R^2, expected accuracy >85%

Validation

Compare ARIMA, LSTM, Hybrid models

Data

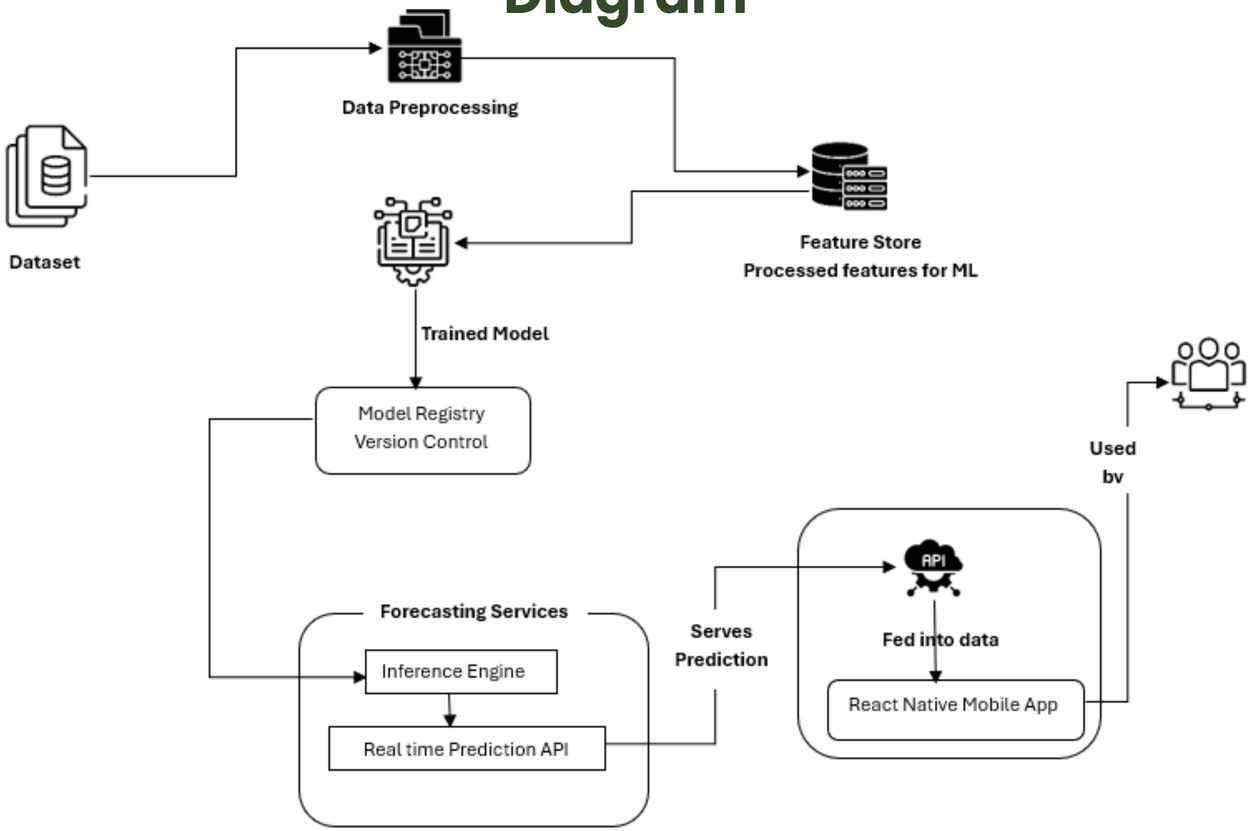
Export records (grades, volumes, export country, shipping costs, demand Index)

Ethical Clearance

Industry-level trade data, no personal data ethically safe

AI - Driven Price Forecasting Grade Specific Ceylon Tea

System Diagram



AI - Driven Price Forecasting Grade Specific Ceylon Tea

Real World Usage

- Exporters, brokers, policy makers use AI - powered forecasts.
- Aligns production and logistics with global demand.
- Reduces risks, improves competitiveness of Ceylon Tea.

Commercialization

1. Investment & Recovery

- Costs: Software, hardware, data collection, marketing
- Recovery: Subscription model, tiered pricing, consulting services

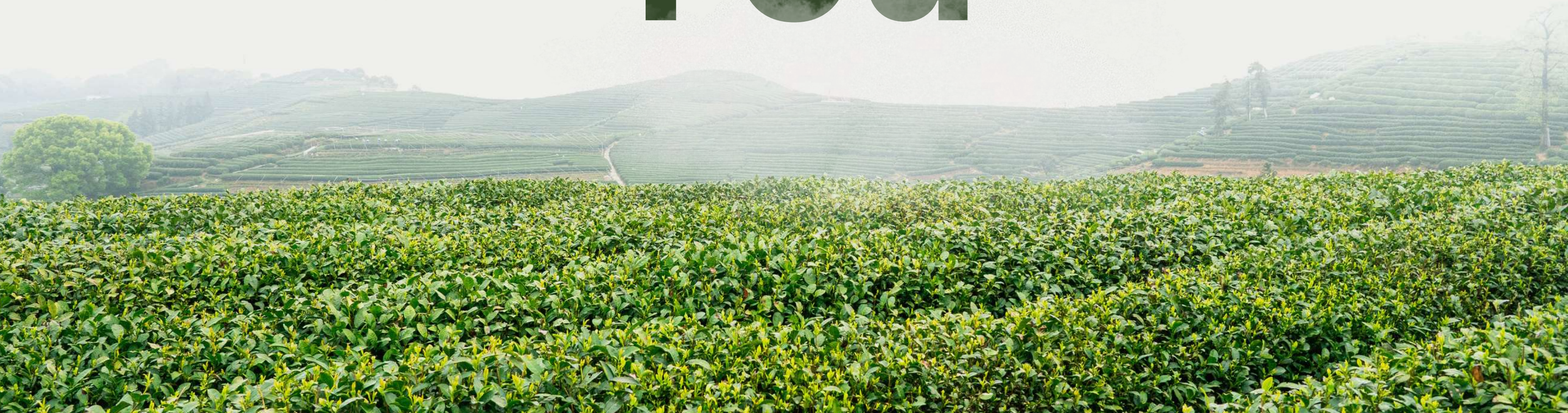
2. **Target Market**

- **Primary:** Smallholder tea estate owners supplying directly to factories.
- **Secondary: 500+ registered tea factories and collection centers across Sri Lanka**
- **Global Expansion:** Other tea-producing countries such as India, Kenya, and China.

3. Value Proposition

- Early disease detection → reduces crop losses
- Harvest prediction & labor cost estimation → optimizes workforce
- Tea quality classification → ensures premium-grade tea
- AI price forecasting → supports data-driven pricing decisions

4. Revenue Models

- **Licensing Model**: One-time license or annual subscription for factories.
- **Enterprise Partnerships:** Collaboration with the Sri Lanka Tea Board

Thank You